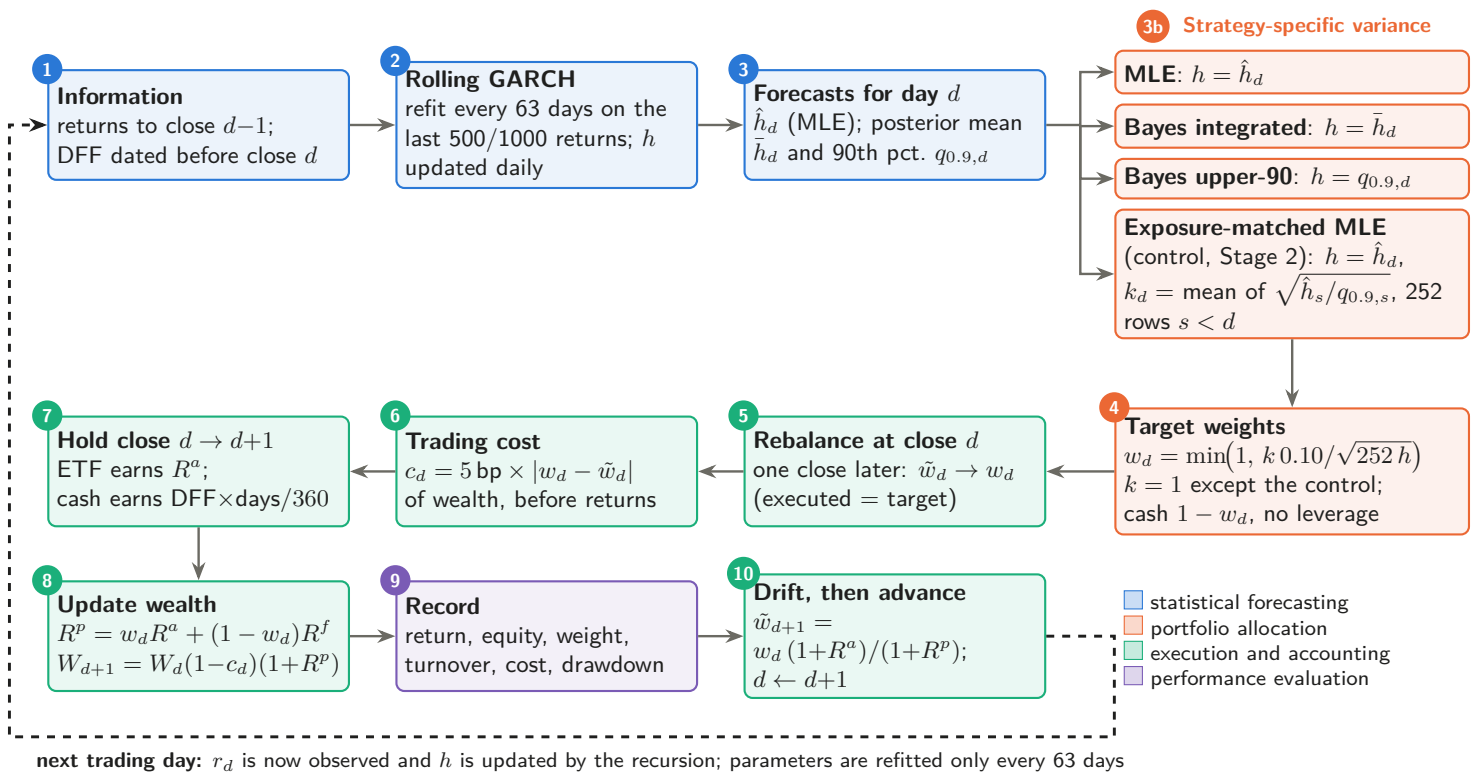

1
Information
returns to close $d-1$;
DFF dated before close d
2
Rolling GARCH
refit every 63 days on the last 500/1000 returns; h updated daily
3
Forecasts for day d
$\hat{h}_d$ (MLE); posterior mean $\bar{h}_d$ and 90th pct. $q_{0.9,d}$
3b
Strategy-specific variance
MLE: $h = \hat{h}_d$
Bayes integrated: $h = \bar{h}_d$
Bayes upper-90: $h = q_{0.9,d}$
Exposure-matched MLE
(control, Stage 2): $h = \hat{h}_d$, k_d = mean of $\sqrt{\hat{h}_s / q_{0.9,s}}$, 252 rows $s < d$
4
Target weights
$w_d = \min\left(1,\ k\,0.10/\sqrt{252\,h}\right)$
$k = 1$ except the control;
cash $1 - w_d$, no leverage
5
Rebalance at close d
one close later: $\tilde{w}_d \to w_d$
(executed = target)
6
Trading cost
$c_d = 5$ bp $\times\ |w_d - \tilde{w}_d|$
of wealth, before returns
7
Hold close $d \to d+1$
ETF earns R^a;
cash earns DFF×days/360
8
Update wealth
$R^p = w_d R^a + (1 - w_d) R^f$
$W_{d+1} = W_d(1-c_d)(1+R^p)$
9
Record
return, equity, weight, turnover, cost, drawdown
10
Drift, then advance
$\tilde{w}_{d+1} =$
$w_d\,(1+R^a)/(1+R^p)$;
$d \leftarrow d+1$
statistical forecasting
portfolio allocation
execution and accounting
performance evaluation
next trading day: r_d is now observed and h is updated by the recursion; parameters are refitted only every 63 days